

CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ SOLUTIONS

Chapter 7: Coordinate Geometry

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 The distance formula

1. $\triangle ABC$ has vertices $A(5, -6)$, $B(6, 4)$ and $C(0, 0)$. Find the length of its median AD . (a) $\sqrt{68}$ units (b) $2\sqrt{15}$ units (c) $\sqrt{101}$ units (d) 10 units **CBSE 2024, 1 mark**

SOLUTION

D is the mid-point of BC : $D = \left(\frac{6+0}{2}, \frac{4+0}{2} \right) = (3, 2)$

$$AD = \sqrt{(5-3)^2 + (-6-2)^2} = \sqrt{4+64} = \sqrt{68}$$

Answer: (a)

2. The points $(3, -5)$ and $(x, -5)$ are 15 units apart. Find the possible values of x . (a) 12, -18 (b) -12, 18 (c) 18, 5 (d) -9, -12 **CBSE 2024, 1 mark**

SOLUTION

Both points have the same y -coordinate, so the distance is $|x - 3| = 15$.

$$x - 3 = 15 \text{ or } x - 3 = -15, \text{ so } x = 18 \text{ or } x = -12.$$

Answer: (b)

2 The section formula

3. A circle has centre $(2, 0)$, and one end of a diameter is $(6, 0)$. Find the other end. (a) $(0, 0)$ (b) $(4, 0)$ (c) $(-2, 0)$ (d) $(-6, 0)$ **CBSE 2024, 1 mark**

SOLUTION

The centre is the mid-point of the diameter. Let the other end be (x, y) .

$$\frac{6+x}{2} = 2 \text{ gives } x = -2, \text{ and } \frac{0+y}{2} = 0 \text{ gives } y = 0.$$

Answer: (c)

4. The point $\left(\frac{8}{5}, y\right)$ lies on the segment joining $(1, 2)$ and $(2, 3)$. In what ratio does it divide the segment? Find y as well. **CBSE 2024, 3 marks**

SOLUTION

Let the ratio be $k : 1$. Then $\frac{2k + 1}{k + 1} = \frac{8}{5}$.

$10k + 5 = 8k + 8$, so $k = \frac{3}{2}$. The ratio is $3 : 2$.

$$y = \frac{3 \times 3 + 2 \times 2}{3 + 2} = \frac{13}{5}$$

Answer: ratio $3 : 2$ and $y = \frac{13}{5}$

5. $A(-1, -1)$, $B(-1, 6)$, $C(3, 6)$ and $D(3, -1)$ are the vertices of a rectangle. P , Q , R and S are the mid-points of AB , BC , CD and DA . Show that the diagonals of $PQRS$ bisect each other. **CBSE 2024, 3 marks**

SOLUTION

$$P = \left(-1, \frac{5}{2}\right), Q = (1, 6), R = \left(3, \frac{5}{2}\right), S = (1, -1)$$

$$\text{Mid-point of diagonal } PR = \left(\frac{-1 + 3}{2}, \frac{5/2 + 5/2}{2}\right) = \left(1, \frac{5}{2}\right)$$

$$\text{Mid-point of diagonal } QS = \left(\frac{1 + 1}{2}, \frac{6 - 1}{2}\right) = \left(1, \frac{5}{2}\right)$$

The diagonals PR and QS have the same mid-point, so they bisect each other.

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